

**NATIONAL UNIVERSITY OF MODERN
LANGUAGE ISLAMABAD**

Assignment:-03



Submitted to:

Sir. Waqar

Submitted by:

Zulqarnain Hassan

Roll No: SP21379

Subject: Artificial Intelligence

Department: BSSE-4-Afternoon

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Task: Answer the following practical questions related to machine learning algorithms:

1. **K-Nearest Neighbors (KNN):** Given the following dataset of house sizes (in square feet) and prices (in \$1000), use the K-Nearest Neighbors algorithm to predict the price of a house with a size of 2200 sq ft, where $k=3$.

- Dataset:

Size (sq ft) | Price (in \$1000)

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1000 | 200

1500 | 300

1800 | 350

2000 | 400

2500 | 500

Answer:

Problem Statement: Predict the price of a house with a size of 2200 sq ft using KNN where $k=3$.

Dataset:

Size (sq ft)	Price (in \$1000)
1000	200
1500	300
1800	350
2000	400
2500	500

Implementation:

1. **Calculate Euclidean Distance:**
 - Formula:
 - Compute distances from 2200 to all other sizes:
 - $2200 - 1000 = 1200$
 - $2200 - 1500 = 700$
 - $2200 - 1800 = 400$
 - $2200 - 2000 = 200$
 - $2200 - 2500 = 300$
2. **Identify 3 Nearest Neighbors:**
 - Closest points are 2000, 1800, and 2500.
3. **Predict Price:**
 - Average price of neighbors:

2. **Naive Bayes Classifier:** Given the following dataset of weather conditions and whether people went to the park (yes/no), implement a Naive Bayes classifier to predict if a person will go to the park based on weather conditions:

Weather | Temperature | Humidity | Go to Park

-----|-----|-----|-----

Sunny | Hot | High | No

Rainy | Mild | Low | Yes

Cloudy | Mild | Low | Yes

Sunny | Cold | Low | Yes

Rainy | Hot | High | No

Task:

- Calculate the prior probabilities $P(\text{Yes})$ and $P(\text{No})$.
- Calculate the likelihoods $P(\text{Weather}|\text{Yes})$, $P(\text{Temperature}|\text{Yes})$, etc.).
- Use Bayes' theorem to predict the class label (Yes/No) for a new instance with the following features: Weather=Sunny, Temperature=Hot, Humidity=Low.

Answer:

Dataset:

Weather	Temperature	Humidity	Go to Park
Sunny	Hot	High	No
Rainy	Mild	Low	Yes
Cloudy	Mild	Low	Yes
Sunny	Cold	Low	Yes
Rainy	Hot	High	No

New Instance: Weather=Sunny, Temperature=Hot, Humidity=Low

Implementation:

Step 1: Calculate Prior Probabilities:

$P(\text{Yes}) = \text{Number of 'Yes'} / \text{Total instances} = 3/5 = 0.6$

$P(\text{No}) = \text{Number of 'No'} / \text{Total instances} = 2/5 = 0.4$

Step 2: Calculate Likelihoods:

$P(\text{Weather}=\text{Sunny} | \text{Yes}) = \text{Sunny with Yes} / \text{Total 'Yes'} = 1/3$

$P(\text{Weather}=\text{Sunny} \mid \text{No}) = \text{Sunny with No} / \text{Total 'No'} = 1/2$

$P(\text{Temperature}=\text{Hot} \mid \text{Yes}) = \text{Hot with Yes} / \text{Total 'Yes'} = 1/3$

$P(\text{Temperature}=\text{Hot} \mid \text{No}) = \text{Hot with No} / \text{Total 'No'} = 1/2$

$P(\text{Humidity}=\text{Low} \mid \text{Yes}) = \text{Low with Yes} / \text{Total 'Yes'} = 2/3$

$P(\text{Humidity}=\text{Low} \mid \text{No}) = \text{Low with No} / \text{Total 'No'} = 0$

Step 3: Apply Bayes' Theorem

For "Yes":

$P(\text{Yes} \mid \text{features}) = P(\text{Yes}) \cdot P(\text{Weather}=\text{Sunny} \mid \text{Yes}) \cdot P(\text{Temperature}=\text{Hot} \mid \text{Yes}) \cdot P(\text{Humidity}=\text{Low} \mid \text{Yes})$

$P(\text{Yes} \mid \text{features}) = 0.6 \cdot 1/3 \cdot 1/3 \cdot 2/3 = 0.6 \cdot 2/27 = 0.0444$

For "No":

$P(\text{No} \mid \text{features}) = P(\text{No}) \cdot P(\text{Weather}=\text{Sunny} \mid \text{No}) \cdot P(\text{Temperature}=\text{Hot} \mid \text{No}) \cdot P(\text{Humidity}=\text{Low} \mid \text{No})$

$P(\text{No} \mid \text{features}) = 0.4 \cdot 1/2 \cdot 1/2 \cdot 0 = 0$

Step 4: Result:

Since $P(\text{Yes} \mid \text{features}) > P(\text{No} \mid \text{features})$ classify the instance as **"Yes."**

3. **K-Means Clustering:** Given the following 2D dataset with coordinates (x, y), perform k-means clustering with k=2:

- **Dataset:**

(2, 3), (3, 4), (6, 8), (7, 9), (2, 1), (8, 8)

Task:

- Step 1: Randomly initialize two centroids.
- Step 2: Assign each point to the closest centroid.
- Step 3: Recalculate the centroids and repeat the assignment until convergence.
- Show the final clusters and the positions of the centroids.

Answer:

Dataset:

Point Coordinates (x, y)

- | | |
|---|--------|
| 1 | (2, 3) |
| 2 | (3, 4) |
| 3 | (6, 8) |
| 4 | (7, 9) |
| 5 | (2, 1) |
| 6 | (8, 8) |

Implementation:

1. **Initialize Centroids:** Randomly choose two points.
2. **Assign Points to Closest Centroid:** Calculate distances and assign clusters.
3. **Recalculate Centroids:** Average the points in each cluster.
4. **Repeat Until Convergence:** Stop when centroids stabilize.

Final Output:

- Cluster 1: Points (2, 3), (3, 4), (2, 1)
- Cluster 2: Points (6, 8), (7, 9), (8, 8)
- Centroids: (2.33, 2.67) and (7, 8.33)